Project Charter

Project Name: Waste Recycling

Project Sponsor: Ministry of the Environment.

Project Manager: Eng/Mohammed Salah

Team members:

Shady Abdelfattah Mohammed

Abd El Aziz Abdelrahman

Mostafa Atyya Ahmed

Sief El Dien Essam

Santy Osama

Shrouk Sobhi

Purpose of Project:

Help reduce waste and achieve the greatest possible benefit, maintain a healthy environment and clean air as much as possible, in addition to reducing epidemics and diseases resulting from the accumulation of waste.

Business Case:

With the increase in the population and the increase in the number of consumed waste and as a result of dumping garbage in the streets, this leads to many problems, including pollution that affects the environment in addition to diseases, epidemics and many other difficulties.

The goal of this project is to create a prototype of a waste recycling system, a mobile application, and a website known as the smart recycling system.

This system will be used by any area where there is garbage, the system will help in being able to recycle and benefit from the garbage as well as protect the environment and the population from the accumulation of waste.

Project Scope:

1- The application allows administrators to access the place of accumulation of garbage.

2- The system will help classify the used garbage to get the most benefit from it.

3- The website will allow users to report any place where the amount of waste increases.

Key Deliverables:

Waste Recycling mobile application that will work on both Android and IOS.

Project Milestones:

• Requirements Review: Requirement's specifications are complete, correct, approved and suitable. By 1/1/2022

• Critical Design Review: Detailed designs fully implement the system architecture, are approved and are suitable for input into the development of code. By 1/3/2022.

• System Test Review: The software product has passed system testing and is suitable for input into acceptance testing.by 1/6/2022.

• Product Operational: The software is in use in its target operational environment.by 1/9/2022.

Project Resources:

• Staff: 2 android developers ,2 IOS developers, 1 Full Stack Web-Developer.

• Hardware: 5 Laptop Devices, Server

• Software: Software licenses (Editors: Android-Studio, App Code, PHP Storm)

Budget:

• 500,000 L.E

Constraints:

• This project must be delivered within the specified budget.

• This project must be delivered on 1/1/2023.

Assumptions (required event must be occur every fixed period):

• Will-get all resources required.

• All-important stakeholders will come to the specified meetings. • All team members continue to the end of the project.

High Level Risks:

• Inaccurate estimates.

• Project team misunderstand requirements.

• High System maintenance costs.

• Loss of key personnel.

• Customer will not accept the software as delivered even though it meets all specifications.

• Selected technology is a poor match to the problem or customer.

• Projects within the program often need the same resources at the same time.

• Server Crashes.

Project Scope

Official Project Name: Smart Recycling System

The Project Scope statement defines these items:

- **Product Scope Description:**

Our project will create a recycling prototype, mobile app and website known as Smart Recycling System. This system will be used by any area with a large population. The system will help control the recycling of waste directly and also allow the waste to be classified to benefit

more from it. The classification will be based on whether the waste is organic or not.

The mobile application will help Administrators to see where waste accumulates to be recycled.

The website, which can be accessed anywhere around it, will be used to find out where the recycling bins are located.

- **Product Acceptance Criteria:**

Requirements Review:

Requirement's specifications are complete, correct, approved and suitable. By 1/1/2022

- **Critical design review:**

Detailed designs fully implement the system architecture, are approved and are suitable for input into the development of code. By 1/3/2021

- **System test Review:**

The software product has passed system testing and is suitable for input into acceptance testing.by 1/9/2022

- **Testing levels:**

1. Unit Testing: Test individual component.
2. Integration Testing: Test Integrated-Component.
3. System Testing: Test entire System.
4. Acceptance Testing: Test The final System.

- **Product operational:**

The software is in use in its target operational environment.by

1/12/2022

- All of sponsors agree about project documents (charter, scope) …

• Project Deliverables:

Inatiation	Planning	Execution & Controlling	Closing
• project charter • project scope	• client observation • Designing • Strategic report • interviwes & meetings • Marketing study	• Design review • Product prototype • Progress report • Execution process improvement • testing Product quality • Improved process efficiency	• Completed product (mobile application and a website)

• Project Exclusions:

Anything like devices, software, hardware that Go beyond the budget are out of scope, and anything that develop the project are in scope.

• Android & IOS new features support.

• System will be usable in center cities.

• Project Constraints:

• This Project must be delivered within the specified budget (hinted at charter).

• This Project must be delivered on 1/1/2023.

• Project Assumptions:

We hope our project to achieve its goals.

• will get all resources required.

• All-important stakeholders will come to the specified meetings.

• All team members continue to the end of the project.

WBS & WBS Dictionary

1.1. Software Development:

Organized by: Team 1

Cost: 100,000 L.E

Milestones: For Three months, the market will be built and designed.

1.1.1 Designing.

1.1.2 Find and buy a suitable place.

1.1.3 Legal requirements.

1.2. Hardware and Security:

Organized by: Team 2

Cost: 200,000 L.E

Milestones: For One month a team of technical engineers will set the sensors and the cameras.

1.2.1 Reliable network.

1.2.1.1 Router.

1.2.1.2 Switches.

1.2.1.3 LAN.

1.2.1.4 WAN.

1.2.2 Cameras

1.2.3 Sensors.

1.3. Software:

Organized by: Team 3

Cost: 200,000 L.E

Milestones: For Four months, a team of programmers will make the mobile application.

1.3.1 Planning for the app.

1.3.2 Design the app.

1.3.3 Coding.

1.3.4 Testing the app.

1.3.5 Publish the app.

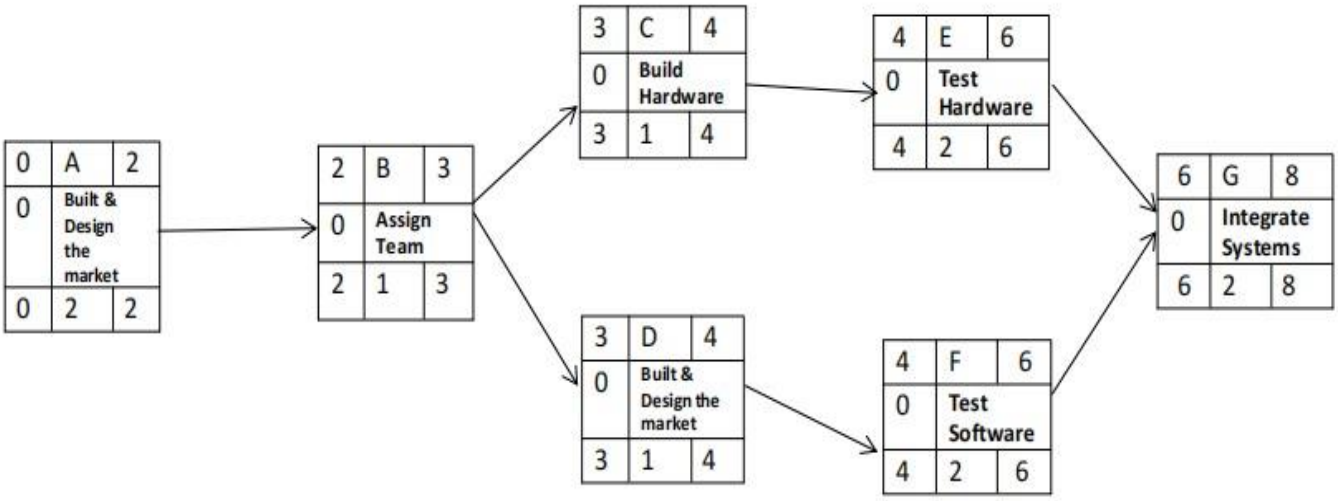
Responsibility Matrix

	Project manager	Backend developer	Frontend developer	Network engineer	Quality control (tester)	Expert	Ads specialist
Analyze the application	R	P	P	P		P	
Write The Code	A	C	C	R		R	
Test the code	A	P	P		C	R	
Make the design	A		C			R	
Package the application	R			P		P	
Test the application release	R				C	P	
Push the application	A	P	P	P		C	
marketing	A					R	C

A = Approves, P = Participant, C = Creator, R = Reviews

Activity-on-Node Network:

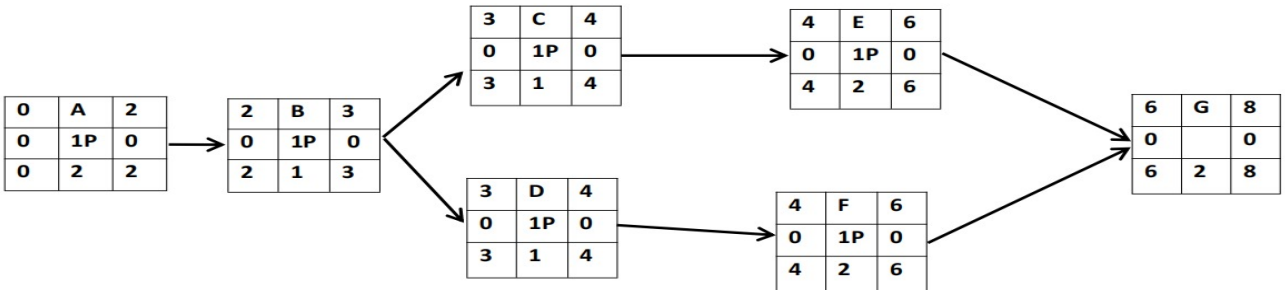
Activity	Description	Preceding Activity	Activity Time
A	Built & Design the market	None	2 workmothes
B	Assign Team	A	1
C	Build Hardware	B	1
D	Code Software	B	1
E	Test Hardware	C	2
F	Test Software	D	2
G	Integrate Systems	E & F	2



ES	ID	EF
SL	RES	SL
LS	DUR	LF

Resources:

ID	RES	DUR	ES	LF	SL	0	1	2	3	4	5	6	7	8
A	1P	2	0	2	0	1	1							
B	1P	1	2	3	0			1						
C	1P	1	3	4	0				1					
D	1P	1	3	4	0					1				
E	1P	2	4	6	0						1	1		
F	1P	2	4	6	0						1	1		
G	1P	2	6	8	0								1	1
Total resource load						1p	1p	1p	1p	1p	2p	2p	1p	1p
Resource available						2p	2p	2p	2p	2p	2p	2p	2p	3p

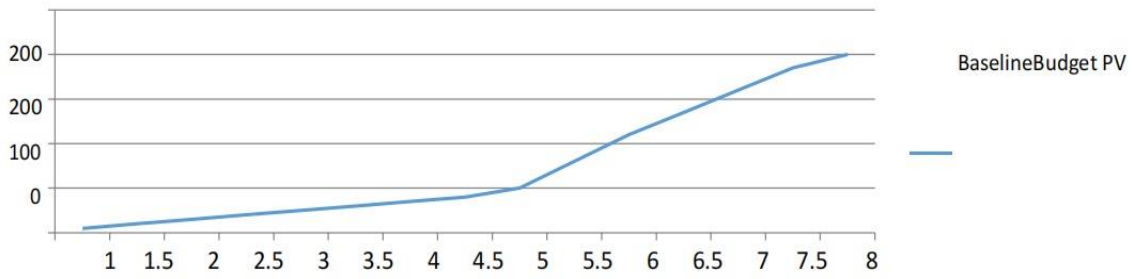


Planned Value (PV):

ID	0	1	2	3	4	5	6	7	8
A	50	50							
B			50						
C				50					
D					50				
E						25	25		
F						50	50		
G								25	50

ID	DUR	Task	Budget	0	1	2	3	4	5	6	7	8
A	8	Built & Design the market	100	50	50							
B	1	Assign Team	50			50						
C	2	Build Hardware	50				50					
D	2	Code Software	50					50				
E	3	Test Hardware	50						25	25		
F	3	Test Software	100						50	50		
G	1	Integrate Systems	100								50	50
Week total			500$	50	50	50	50	50	75	75	50	50
cumulative				50	100	150	200	250	325	400	450	500

Baseline Budget PV:



Risk Management

1-The Risks That can face our Project:

- Increase in the cost.
- Not to satisfy the users of product.
- Problems in application's system.
- Security Hack.

2-How to deal with these Risks?

- Increase in the cost Reducing costs and using less expensive materials.
- Not to satisfy the users of produce Create feedback on the application to hear customer issues and try to solve them and make offers and discounts.
- Problems in application's system Hiring Technician team to ensure that the application runs well.
- Security Hack Improve the security of the app and use encryption.

3-Monitor the Project:

- Creating milestones.
- Set realistic, clear and measurable goals.
- Meet regularly with team and stakeholders.
- Have clear deadlines.
- Track Key Performance.